

BSDS (2025 Semester II): Statistics II

Endsem: Part B

NAME :

ROLL :

Time: 90 minutes

Total marks: 65

Maximum attainable marks: 55

Answer all questions.

1. Let X_n , $n \geq 1$ be a sequence of random variables of the following form:

$$P(X_n = k) = \left(1 - \frac{1}{n}\right) \frac{1}{N}, \quad \text{for } k = 1, \dots, N, \quad \text{and} \quad P(X_n = \sqrt{n}) = \frac{1}{n}.$$

Show that $X_n \xrightarrow{d} Y$ as $n \rightarrow \infty$, where Y is distributed as a discrete uniform distribution supported on $\{1, \dots, N\}$. Without loss of generality, you may assume $n > N^2$. [10]

[Hint: You may solve using the following steps:

- Derive distribution functions of X_n and Y , say, F_n and F_Y , respectively.
- Identify the set of discontinuity points of F_Y , say $\mathcal{D}$.
- For each $y \in \mathbb{R} \setminus \mathcal{D}$, show that $F_n(y) \rightarrow F_Y(y)$ as $n \rightarrow \infty$.]

Let F_n and G be the CDFs of X_n and Y , respectively, $n \gg 1$.

Observe that,

$$F_n(x) = \begin{cases} 0 & \text{if } x < 1 \\ \left(1 - \frac{1}{n}\right) \frac{1}{N} & \text{if } 1 \leq x < 2 \\ \left(1 - \frac{1}{n}\right) \frac{2}{N} & \text{if } 2 \leq x < 3 \\ \vdots & \\ \left(1 - \frac{1}{n}\right) \frac{N-1}{N} & \text{if } N-1 \leq x < N \\ \left(1 - \frac{1}{n}\right) & \text{if } N \leq x < \sqrt{n} \\ 1 & \text{if } x \geq \sqrt{n} \end{cases}$$

Further,

$$G(x) = \begin{cases} 0 & \text{if } x < 1 \\ \frac{1}{N} & \text{if } 1 \leq x < 2 \\ \vdots & \\ \frac{N-1}{N} & \text{if } N-1 \leq x < N \\ 1 & \text{if } x \geq N \end{cases}$$

Clearly, the ^{set of} points of discontinuities of G is $\mathcal{D}_G = \{1, \dots, N\}$.

To show, $X_n \xrightarrow{d} Y$, we need to show that

$$(*) - \quad F_n(x) \rightarrow G(x) \quad \text{for all } x \in \mathbb{R} \setminus \mathcal{D}_G, \text{ as } n \rightarrow \infty.$$

Let $x \in (-\infty, 1)$ then $F_n(x) = G(x) = 0 \quad \forall n$.

$x \in (1, 2)$ then $F_n(x) = \left(1 - \frac{1}{n}\right) \frac{1}{N} \rightarrow \frac{1}{N} = G(x)$ as $n \rightarrow \infty$.

In general, if $x \in (K-1, K)$ for $K=2, \dots, N$ then

$$F_n(x) = \left(1 - \frac{1}{n}\right) \frac{K-1}{N} \rightarrow \frac{K-1}{N} = G(x) \text{ as } n \rightarrow \infty.$$

Next, let $x \in (N, \sqrt{n})$ then $F_n(x) = \left(1 - \frac{1}{n}\right) \rightarrow 1 = G(x)$ as $n \rightarrow \infty$,

and for $x \geq \sqrt{n}$, $F_n(x) = G(x) = 1 \quad \forall n$.

Hence, $(*)$ is proved, and $X_n \xrightarrow{d} Y$ as $n \rightarrow \infty$.

2. Let $X_1, \dots, X_n$ be i.i.d. random variables from $\text{uniform}(0, 1)$ distribution. Find the distribution of $Y_n = \sum_{i=1}^n \log(1/X_i)$.

[You may refer to the MGF table provided in the back of the question-answer booklet.] [7]

Let $Z_i = -\log X_i = \log(1/X_i)$ for $i=1, \dots, n$.

$$\begin{aligned} \text{MGF of } Z, \quad M_Z(t) &= E(e^{tZ}) = E(e^{-t \log X}) = E(X^{-t}) \\ &= \int_0^1 x^{-t} dx = \frac{x^{1-t}}{1-t} \Big|_0^1 = \frac{1}{(1-t)} \quad \text{for } t < 1 \end{aligned}$$

Further, $Z_1, \dots, Z_n$ are i.i.d. as $X_1, \dots, X_n$ are i.i.d.

So, the MGF of Y_n is:

$$\begin{aligned} M_{Y_n}(t) &= E(e^{tY_n}) = E(e^{t \sum_{i=1}^n Z_i}) = \prod_{i=1}^n E(e^{tZ_i}) \\ &= [M_Z(t)]^n \\ &= (1-t)^{-n}, \quad t < 1. \end{aligned}$$

It is given that if $W \sim \text{Gamma}(\alpha, \beta)$ then $M_W(t) = (1-\beta t)^{-\alpha}$ for $t < 1/\beta$.

Thus, by uniqueness of MGF we have $Y_n \sim \text{Gamma}(n, 1)$.

3. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{location-exponential}(\alpha, \theta_0)$, with p.d.f.

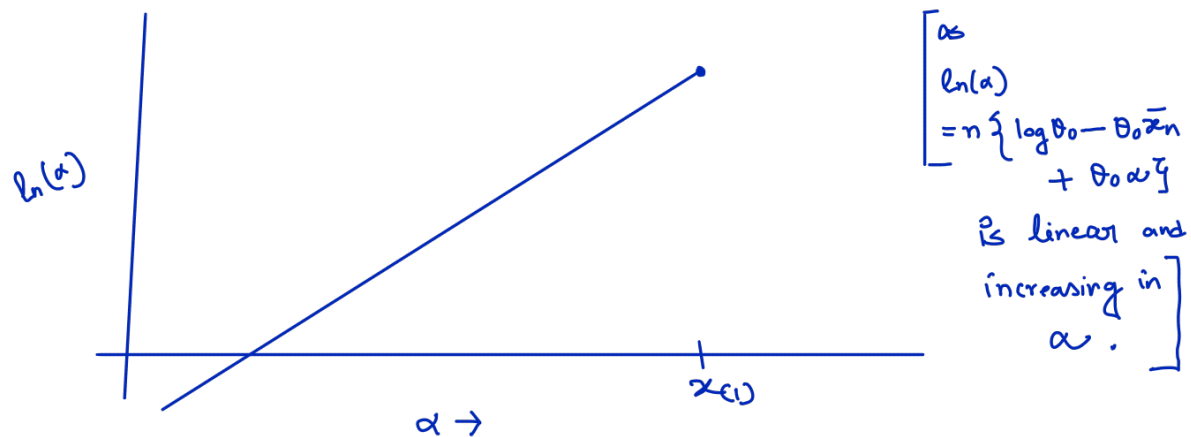
$$f_X(x) = \theta_0 \exp\{-\theta_0(x - \alpha)\}, \quad x \geq \alpha, \quad \theta_0 > 0 \text{ is known, } \alpha \in \mathbb{R}.$$

- Find the MLE of α .
- Given that the MLE of α is a complete and sufficient statistics (CSS), can you find the UMVUE of α ?
- If the MLE is different from the UMVUE, then identify the better estimator among these two, by comparing their MSEs. [5 + 6 + 4]

(a) The log-likelihood function of α is

$$\ln(\alpha) = \begin{cases} n \log \theta_0 - \theta_0 \sum_{i=1}^n (x_i - \alpha) & \text{if } x_i \geq \alpha \quad \forall i=1, \dots, n \\ & \text{i.e., if } x_{(1)} \geq \alpha. \\ -\infty & \text{if } x_{(1)} < \alpha. \end{cases}$$

Thus, the graph of $\ln(\alpha)$ w.r.t. α is



Hence, $\ln(\alpha)$ is maximized at $x_{(1)}$.

Thus, $\hat{\alpha}_{MLE} = X_{(1)}$.

(b) Given that $X_{(1)}$ is complete and sufficient (CSS) for α , let us find the distribution and expectation of $X_{(1)}$.

Let G_n and g_n be the CDF and PDF of $T_n = X_{(1)}$.

For $t > \alpha$;

$$G_n(t) = P(T_n \leq t) = 1 - [P(X_1 > t)]^n$$

$$= 1 - \left[\int_t^{\infty} \theta_0 e^{-\theta_0(x-\alpha)} dx \right]^n$$

Define

$$u = \theta_0(x-\alpha)$$

$$du = \theta_0 dx$$

x	t	∞
u	$\theta_0(t-\alpha)$	∞

$$= 1 - \left[\int_{\theta_0(t-\alpha)}^{\infty} \bar{e}^u du \right]^n = 1 - \bar{e}^{-n\theta_0(t-\alpha)}$$

$$\text{Thus, } g_n(t) = \begin{cases} n\theta_0 \bar{e}^{-n\theta_0(t-\alpha)} & \text{for } t > \alpha \\ 0 & \text{otherwise.} \end{cases}$$

Finally,

$$E(T_n) = E(X_{(1)}) = n\theta_0 \int_{\alpha}^{\infty} \bar{e}^{-n\theta_0(t-\alpha)} t dt$$

$$= \int_0^{\infty} \bar{e}^{-u} \left(\alpha + \frac{u}{n\theta_0} \right) du$$

where

$$u = n\theta_0(t-\alpha)$$

$$du = n\theta_0 dt$$

$$= \alpha \int_0^{\infty} \bar{e}^{-u} du + \frac{1}{n\theta_0} \int_0^{\infty} u \bar{e}^{-u} du$$

t	α	∞
u	0	∞

$$= \alpha [1] + \frac{1}{n\theta_0} [2]$$

$$t = \alpha + \frac{u}{n\theta_0}$$

$$= \alpha + \frac{1}{n\theta_0}$$

Thus, $E\left(T_n - \frac{1}{n\theta_0}\right) = \alpha$. Since $\left(T_n - \frac{1}{n\theta_0}\right)$ is a function of CSS, and is unbiased for α , it must be the UMVUE of α .

$$\text{So, } \hat{\alpha}_{\text{UMVUE}} = T_n - \frac{1}{n\theta_0} = X_{(1)} - \frac{1}{n\theta_0}$$

$$(c) \quad \hat{\alpha}_{\text{MLE}} = X_{(1)} \quad \text{and} \quad \hat{\alpha}_{\text{UMVUE}} = X_{(1)} - \frac{1}{n\theta_0}$$

$$\left. \begin{array}{l} \text{Bias of } \hat{\alpha}_{\text{MLE}} \text{ is : } E(X_{(1)}) - \alpha = -\frac{1}{n\theta_0} \\ \text{Bias of } \hat{\alpha}_{\text{UMVUE}} \text{ is } 0. \end{array} \right\} \text{--- (A)}$$

⑥ — Variance of $\hat{\alpha}_{MLE}$ = Variance of $\hat{\alpha}_{UMVUE}$,
 as $\text{var}\left(X_{(1)} - \frac{1}{n\theta_0}\right) = \text{var}(X_{(1)})$.

Since $MSE = \text{Bias}^2 + \text{variance}$, from ④ and ⑥, we have

$$MSE(\hat{\alpha}_{MLE}) > MSE(\hat{\alpha}_{UMVUE}) \text{ and the difference in MSE is } 1/(n\theta_0)^2.$$

Thus, $\hat{\alpha}_{UMVUE}$ is better estimate than $\hat{\alpha}_{MLE}$ in terms of MSE. However, if n is large, both of them are equivalent.

4. Let $X_1, \dots, X_n, n = 1, 2, \dots$, be i.i.d. samples from F_μ with mean μ and standard deviation $\sigma = 3$. We want to set a confidence interval for μ of the form $[\bar{X}_n - 0.3, \bar{X}_n + 0.3]$.

- (a) Using CLT find the minimum sample size n such that the confidence coefficient is at least 0.95. [Refer to the normal table provided at the end of the question-answer booklet.]
 (b) If you use the Chebyshev inequality instead, what would be the minimum sample size? [6 + 5]

(a) Need to find the minimal sample size n_0 such that for $n \geq n_0$,

$$P(\bar{X}_n - 0.3 \leq \mu \leq \bar{X}_n + 0.3) \geq 0.95, \text{ using CLT.}$$

By CLT we have

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{3} \xrightarrow{d} N(0,1) \text{ as } n \rightarrow \infty.$$

$$\text{Thus, } P(\bar{X}_n - 0.3 \leq \mu \leq \bar{X}_n + 0.3)$$

$$= P(-0.3 \leq \bar{X}_n - \mu \leq 0.3)$$

$$= P\left(-\frac{\sqrt{n} \times 0.3}{3} \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq \frac{\sqrt{n} \times 0.3}{3}\right)$$

$$\approx \Phi\left(\frac{\sqrt{n}}{10}\right) - \Phi\left(-\frac{\sqrt{n}}{10}\right) = 2\Phi\left(\frac{\sqrt{n}}{10}\right) - 1$$

By the criterion we require n_0 s.t. for $n \geq n_0$,

$$2\Phi\left(\frac{\sqrt{n}}{10}\right) - 1 \geq 0.95$$

$$\Rightarrow \Phi\left(\frac{\sqrt{n}}{10}\right) \geq \frac{1.95}{2} = 0.975$$

From the table we have

$$\Phi(1.96) = 0.975$$

$$\text{Thus, } n_0 \text{ is such that } \frac{\sqrt{n_0}}{10} \geq 1.96$$

$$\Rightarrow n_0 \geq (19.6)^2 = 384.16$$

So, the minimal sample size is $n_0 = 385$.

(b) We know that $E(\bar{x}_n) = \mu$ and $\text{var}(\bar{x}_n) = \frac{\sigma^2}{n} = \frac{9}{n}$.

By Chebyshev's inequality

$$\begin{aligned} & P(\bar{x}_n - 0.3 \leq \mu \leq \bar{x}_n + 0.3) \\ &= P(-0.3 \leq \bar{x}_n - \mu \leq 0.3) \\ &= 1 - P(|\bar{x}_n - \mu| > 0.3) \\ &\geq 1 - \frac{\text{var}(\bar{x}_n)}{(0.3)^2} = 1 - \frac{9 \times 100}{n \times 9} = 1 - \frac{100}{n}. \end{aligned}$$

We require the prob. at least 0.95. So the optimal n_0

$$\begin{aligned} \text{satisfies, } 1 - \frac{100}{n_0} &\geq 0.95 \Rightarrow 0.05 \times n_0 \geq 100 \\ &\Rightarrow n_0 \geq \frac{100 \times 100}{5} = 2000 \end{aligned}$$

$\therefore$ Required minimal sample size is $\boxed{n_0 = 2000.}$

5. Consider the double exponential distribution with parameter θ and pdf

$$f_X(x; \theta) = (2\theta)^{-1} \exp\{-|x|/\theta\}, \quad x \in \mathbb{R}, \quad \theta > 0.$$

Find the UMP size- α test for testing $H_0 : \theta \leq \theta_0$ against $H_A : \theta = \theta_1$ based on **one sample** X , where $\theta_1 > \theta_0$. [14]

[Hint: You may go through the following steps:

- i. Find the MP level- α test ϕ^* for testing $H_0^* : \theta = \theta_0$ against H_A , using Neyman-Pearson's lemma.
- ii. Find the power function of the test ϕ^* $\beta_{\phi^*}(\theta)$, and show that $\beta_{\phi^*}(\theta)$ is monotone in θ .
- iii. Hence, show that the size of the test ϕ^* under H_0 , i.e., $\sup_{\theta \leq \theta_0} \beta_{\phi^*}(\theta)$, is highest at $\theta = \theta_0$ and is α .]

i. Let $H_0^* : \theta = \theta_0$ and $H_A : \theta = \theta_1$ where $\theta_1 > \theta_0$

By NP lemma the MP test is of the form

$$\phi^*(x) = \begin{cases} 1 & \text{if } \frac{f_X(x; \theta_1)}{f_X(x; \theta_0)} = \lambda(x) > k \\ 0 & \text{if } \lambda(x) < k, \end{cases}$$

and it satisfies

$$E_{H_0^*}(\phi^*(x)) = \alpha.$$

$$\begin{aligned} \text{Here, } \lambda(x) &= \frac{(2\theta_1)^{-1} \exp\{-|x|/\theta_1\}}{(2\theta_0)^{-1} \exp\{-|x|/\theta_0\}} \\ &= \frac{\theta_0}{\theta_1} \cdot \exp\left\{|x| \left(\frac{1}{\theta_0} - \frac{1}{\theta_1}\right)\right\} \end{aligned}$$

$$\text{Thus, } \lambda(x) > k \Leftrightarrow |x| > c_0 = \left\{ \log k - \log\left(\frac{\theta_0}{\theta_1}\right) \right\} / \left(\frac{1}{\theta_0} - \frac{1}{\theta_1}\right)$$

$$\text{So, } \phi^*(x) = \begin{cases} 1 & \text{if } |x| > c_0 \\ 0 & \text{if } |x| \leq c_0 \end{cases}, \quad \text{as } P_{H_0}(|x| = c_0) = 0$$

(*) -

To find c_0 , equate

$$\begin{aligned} E_{H_0^*}(\phi^*(x)) &= \alpha \Leftrightarrow P_{H_0^*}(|x| > c_0) = \alpha \\ \Leftrightarrow 1 - P_{\theta_0}(|x| \leq c_0) &= \alpha \Leftrightarrow \int_{-c_0}^{c_0} (2\theta_0)^{-1} e^{-|x|/\theta_0} dx = 1 - \alpha \end{aligned}$$

$$\Leftrightarrow \frac{1}{\theta_0} \int_0^{c_0} e^{-x/\theta_0} dx = \frac{1}{\theta_0} \left[-e^{-u} \right]_0^{c_0/\theta_0} = 1 - \alpha$$

$$\text{So, } 1 - e^{-c_0/\theta_0} = \theta_0(1-\alpha) \Rightarrow -c_0/\theta_0 = \log(1 - \theta_0(1-\alpha))$$

$$\Rightarrow c_0 = -\theta_0 \log(1 - \theta_0(1-\alpha)) \quad \text{--- } (*)$$

Thus, the MP test is of the form $(*)$ with c_0 as in $(*)$.

ii. Power function of θ_0 is

$$\beta_{\phi^*}(\theta) = E_{\theta}(\phi^*(x)) = P_{\theta}(|x| > c_0)$$

$$= 1 - \int_{-c_0}^{c_0} (2\theta)^{-1} e^{-|x|/\theta} dx$$

$$= 1 - \frac{1}{\theta} \left[-e^{-u} \right]_0^{c_0/\theta}$$

$$= 1 - \frac{1}{\theta} \left(1 - \exp\{-c_0/\theta\} \right) \quad \text{--- } (I)$$

Observe that, both $1/\theta$ and $\left(1 - \exp\{-c_0/\theta\} \right)$ are decreasing functions of θ . Being product of decreasing functions

$g(\theta) = \frac{1}{\theta} \left(1 - \exp\{-c_0/\theta\} \right)$ is also a decreasing fn. of θ .

Finally, $\beta_{\phi^*}(\theta) = 1 - g(\theta)$ is an increasing function of θ .

iii. As $\beta_{\phi^*}(\theta)$ is an increasing function of θ ,

$$\sup_{\theta \leq \theta_0} \beta_{\phi^*}(\theta) = \beta_{\phi^*}(\theta_0).$$

Again, $\beta_{\phi^*}(\theta_0) = E_{H_0^*}(\phi^*(x)) = \alpha$ From part i.

Thus, ϕ^* remains a level- α (in fact, size α) test if the

-null hypothesis extends from $H_0^* : \theta = \theta_0$ to $H_0 : \theta \leq \theta_0$.

6. Let X be a Pareto distributed random variable with pdf

$$f_{\theta}(x) = \alpha_0 \frac{\theta^{\alpha_0}}{x^{\alpha_0+1}}, \quad x > \theta > 0,$$

and $\alpha_0 > 0$ is known.

- (a) Find an appropriate pivot $T_X(\theta)$, which is a function of X and θ .
 (b) Find (a, b) such that $P_{\theta}(T_X(\theta) < a) = \alpha/2$ and $P_{\theta}(T_X(\theta) > b) = \alpha/2$.
 (c) Hence, find a 95% confidence interval for θ .

[2 + 3 + 3]

(a) Define $W = \theta/X$ then $\left| \frac{dx}{dw} \right| = J_{x \rightarrow w} = \frac{\theta}{w^2}$ and $\begin{array}{c|c|c} x & \theta & \infty \\ \hline w & 1 & 0 \end{array}$

Thus, $f_W(w) = \alpha_0 w^{\alpha_0} \cdot \frac{w}{\theta} \cdot \frac{\theta}{w^2}; \quad 0 < w < 1$
 $= \alpha_0 w^{\alpha_0-1} \quad \text{for } 0 < w < 1. \quad \text{--- (I)}$

So, the distribution of W is free of θ_0 .

Thus, $W = \frac{\theta}{X}$, being a ^{continuous} monotone function of θ and X ,
 and its distribn. being free of θ ,
 is an appropriate pivot for θ .

$\therefore T_X(\theta) = W = \theta/X$.

(b) To find a s.t. $P(T_X(\theta) = W < a) = \frac{\alpha}{2}$
 $\Rightarrow \alpha_0 \int_0^a w^{\alpha_0-1} dw = w^{\alpha_0} \Big|_0^a = a^{\alpha_0} = \frac{\alpha}{2}$
 $\Rightarrow a = \left(\frac{\alpha}{2} \right)^{1/\alpha_0} \quad \text{--- (II)}$

Further, b is such that

$P(T_X(\theta) = W > b) = \frac{\alpha}{2}$
 $\Rightarrow \alpha_0 \int_b^1 w^{\alpha_0-1} dw = w^{\alpha_0} \Big|_b^1 = 1 - b^{\alpha_0} = \frac{\alpha}{2}$

$$\Rightarrow b = (1 - \alpha/2)^{1/\alpha_0} \quad \text{--- (III)}$$

(c) From part (b) [see (II) and (III)],

$$P_{\theta} \left((\alpha/2)^{1/\alpha_0} \leq T_x(\theta) = \frac{\theta}{\bar{X}} \leq (1 - \alpha/2)^{1/\alpha_0} \right) = 1 - \alpha$$

$$\Rightarrow P_{\theta} \left(\bar{X} (\alpha/2)^{1/\alpha_0} \leq \theta \leq \bar{X} (1 - \alpha/2)^{1/\alpha_0} \right) = 1 - \alpha.$$

Thus, $\left[\bar{X} (\alpha/2)^{1/\alpha_0}, \bar{X} (1 - \alpha/2)^{1/\alpha_0} \right]$ is a $(1 - \alpha)$ -confidence interval for θ .

$$\bar{X} = \bar{x}$$

Additional Information

Distribution	Parameters	MGF $M_X(t)$	Range of t
Normal	$\mu \in \mathbb{R}, \sigma^2 > 0$	$\exp(\mu t + \sigma^2 t^2/2)$	$t \in \mathbb{R}$
Exponential	$\lambda > 0$	$\lambda/(\lambda - t)$	$t < \lambda$
Gamma	$\alpha, \beta > 0$	$(1 - \beta t)^{-\alpha}$	$t < 1/\beta$
Double Exponential	$\mu \in \mathbb{R}, b > 0$	$\exp\{\mu t\}/(1 - b^2 t^2)$	$ t < 1/b$

Table 1: MGF of some common distributions.

z	$\Phi(z)$
0.900	0.816
0.925	0.822
0.950	0.829
0.975	0.835
1.280	0.900
1.440	0.925
1.645	0.950
1.960	0.975

Table 2: Table of standard normal CDF.